

8.1: Compound Angle Formulae:

Applying a rotation matrix $\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$ to vector $\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ rotates the line by angle ϕ.

Hence:

$$\begin{pmatrix} \cos(\theta+\phi) \\ \sin(\theta+\phi) \end{pmatrix} = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = \begin{pmatrix} \cos\phi\cos\theta - \sin\phi\sin\theta \\ \sin\phi\cos\theta + \cos\phi\sin\theta \end{pmatrix}$$

Hence

$\cos(\theta+\phi) = \cos\phi\cos\theta - \sin\theta\sin\phi$

$\sin(\theta+\phi) = \sin\phi\cos\theta + \cos\phi\sin\theta$

Using the fact $\sin(-\alpha) = -\sin\alpha$ and $\cos(-\alpha) = \cos\alpha$

$\Rightarrow \cos(\theta-\phi) = \cos\phi\cos\theta + \sin\phi\sin\theta$

$\Rightarrow \sin(\theta-\phi) = \cos\phi\sin\theta - \sin\phi\cos\theta$

Therefore (with simplification)

$$\Rightarrow \tan(\theta+\phi) = \frac{\tan\theta+\tan\phi}{1-\tan\theta\tan\phi}$$

$$\Rightarrow \tan(\theta-\phi) = \frac{\tan\theta-\tan\phi}{1+\tan\theta\tan\phi}$$

8.2: Double Angle Formulae:

Using trivial mathematics:

$\sin 2\theta = 2\sin\theta\cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

8.3: The form $r\sin(\theta \pm \alpha)$

All functions $a\cos\theta + b\sin\theta$ can be written in a form of:

$r\sin(\theta+\alpha) = r\sin\theta\cos\alpha + r\sin\alpha\cos\theta$ $(a\sin\theta + b\cos\theta)$

$r\sin(\theta-\alpha) = r\sin\theta\cos\alpha - r\sin\alpha\cos\theta$ $(a\sin\theta - b\cos\theta)$

$r\cos(\theta+\alpha) = r\cos\theta\cos\alpha - r\sin\theta\sin\alpha$ $(a\cos\theta - b\sin\theta)$

$r\cos(\theta-\alpha) = r\cos\theta\cos\alpha + r\sin\theta\sin\alpha$ $(a\sin\theta + b\cos\theta)$